AKHIL GHARPUREY
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Portfolio Link: https://akhilghar.github.io

EDUCATION

University of Texas at Austin – Austin, TX **GPA: 3.99/4.00**
B.S. in Mechanical Engineering Honors, Minor in Robotics **August 2023 – May 2027**

- Major Coursework: Feedback Control Systems, Introduction to Robotics, Machine Element Design, Mechatronics and Circuits, Numerical Methods, Materials Processing, Heat Transfer, Linear Algebra, Fluid Mechanics

WORK EXPERIENCE

Bakolas Research Group – Austin, TX **January 2026 – Present**
Undergraduate Research Assistant (PI: Efstathios Bakolas, PhD)

- Developing a real-time trajectory optimization algorithm for autonomous package delivery robots in dynamic environments.
- Implementing GPU-accelerated Model-Predictive Control (MPC) methods with Monte Carlo obstacle prediction to accumulate a sub-5% total collision probability.
- Deploying real-time terrain traversability classification and rendering methods on the robot to ensure satisfaction of no-tip and payload safety constraints throughout all generated trajectories.

Human-Centered Robotics Laboratory – Austin, TX **August 2024 – January 2026**
Undergraduate Research Assistant (PI: Luis Sentis, PhD)

- Led the design and development of a human-centered project funded by an industry grant.
- Developed a bang-bang controller interpreting real-time human electrodermal activity (EDA) data and transmitting PWM signal information through BLE.
- Integrated image decomposition methods and Kalman filtering in OpenCV to detect and track the user's position in real time with <5ms latency according to an infrared (IR) camera and emitter.

FlareX – Austin, TX **September 2025 – December 2025**
System Integration Engineer

- Designed a mechanical antenna mount for stable wireless ground communication during flight of an unmanned aerial vehicle (UAV).
- Developed a catapult launch mechanism to minimize fluctuation in the lift-off attack angle of the UAV.

Texas Guadaloop [UT Hyperloop Organization] – Austin, TX **September 2023 – December 2024**
Vehicle Dynamics Lead **June 2024 – December 2024**

- Led a team of four in designing and manufacturing a bogie suspension chassis for a maglev pod, ensuring a stable configuration during nominal operation.
- Employed ANSYS Static and Transient Structural Finite Element simulations to guarantee the bogie's structural integrity under static and dynamic loads with a minimum safety factor of 2.0.
- Created Failure Mode and Effects Analysis (FMEA) sheets to examine the severity and fixability of all failure scenarios for bogie and braking systems.
- Collaborated with 7 engineering teams to coordinate dynamic system integration with maglev interfaces.

PROJECT EXPERIENCE

Tracker Drone Project – Austin, TX **May 2025 – Present**

- Investigated the design and implementation of an FPV quadcopter drone controlled by following a given object.
- Implemented OpenCV and low-latency (< 5ms) image processing techniques on a Raspberry Pi for real-time object tracking via onboard camera.
- Developed a 14" x 14" lightweight (< 1kg) quadcopter chassis to integrate all electronic quadcopter components.
- Designed a linear quadratic tracking feedback controller in MATLAB for a path-following quadcopter.

TECHNICAL SKILLS

- **Design Software:** Proficient in SolidWorks CAD, ANSYS FEA, and MS Office Suite; familiar with KiCad 9.0 Suite
- **Programming:** Proficient in Python, MATLAB, and Simulink; familiar with C/C++/CUDA, Linux, Git, and Arduino IDE
- **Precision Manufacturing:** Experienced with manual lathe and 3D printer; familiar with manual 3-axis mill